Local Models, Sovereign AI & Agentic Governance

PRESENTER
Miles Thompson · AI Engineer · Cyborg Garden

VENUE
AI Engineering Meetup, Wellington

FORM 2-P · PERSONNEL RECORD

SUBJECT: CREDENTIALS

REF. SA-02/23



@utunga

NAME

Miles Thompson

TE HIKU MEDIA

Papa Reo bilingual speech models, trained and self-hosted

CYBORG GARDEN

Local agents and agentic governance

PRESENTED AT

NeurIPS · ICML · AAAI

CLASSIFICATION: PUBLIC

FILED: WELLINGTON

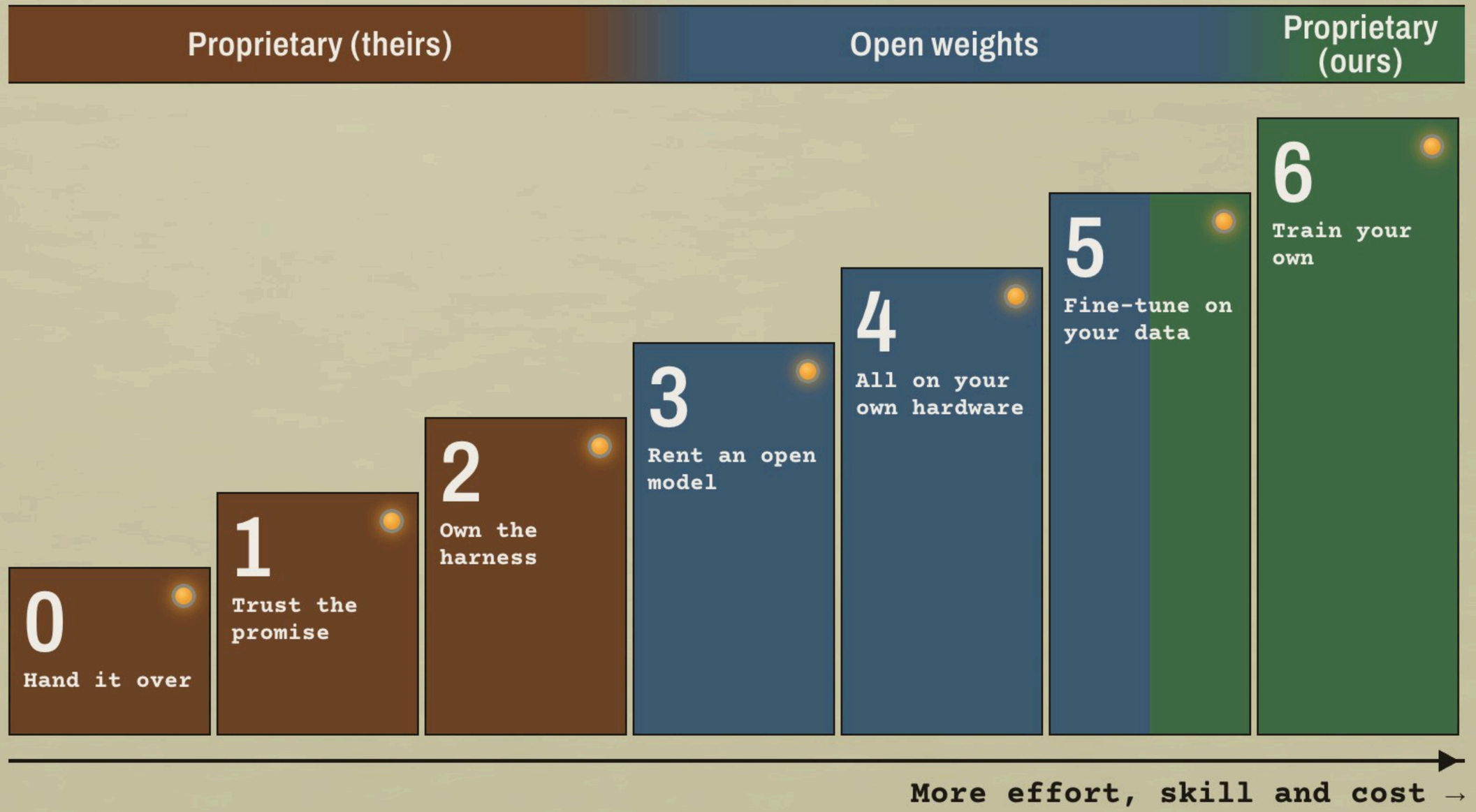
Where does your data go?

2 How do you feel about that?

3 What is the carbon impact?

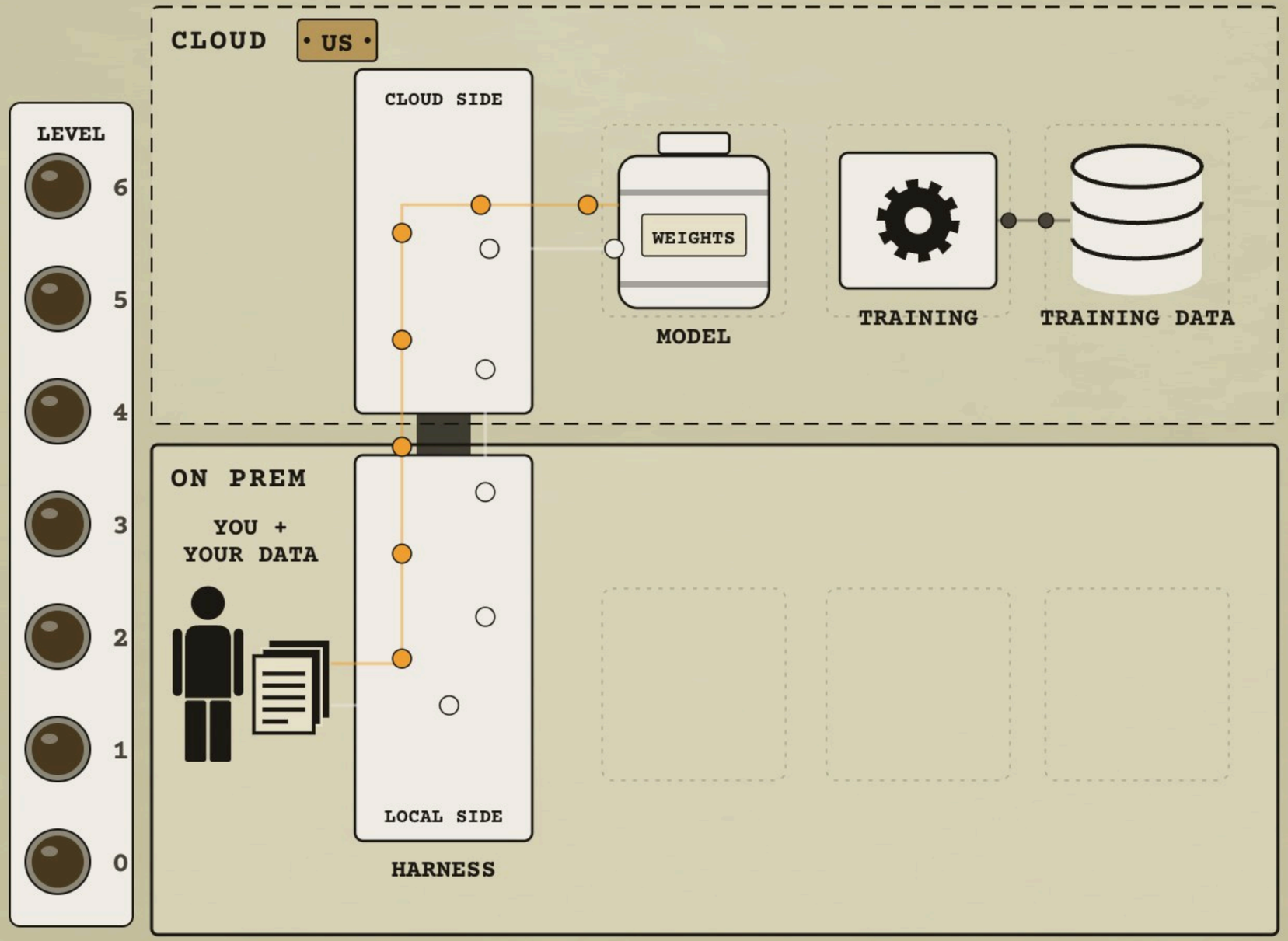
4 Are you losing control and autonomy?

Levels 0-6


Proprietary (theirs)
Open weights
Proprietary (ours)
0
Hand it over
1
Trust the promise
2
Own the harness
3
Rent an open model
4
All on your own hardware
5
Fine-tune on your data
6
Train your own
More effort, skill and cost →

FORM 7-L · LEVELS OF SOVEREIGNTY

VENDOR'S · SOMEONE ELSE'S · YOURS · ON A PROMISE



FORM 7-L · LEVEL REF. SA-08/23

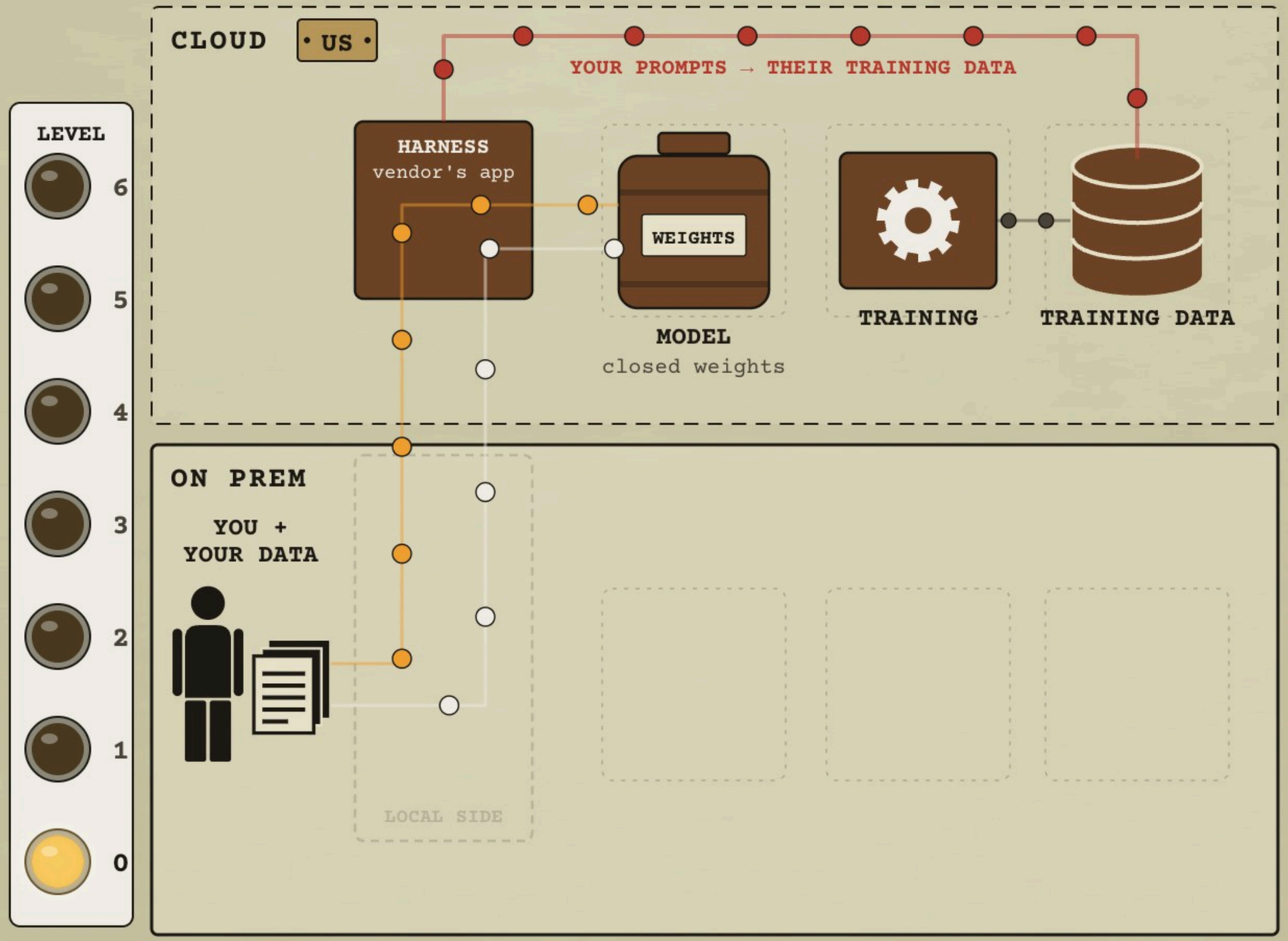
The parts

The parts of an AI system

- Every system has these parts
- Each can be theirs or yours
- Each can be cloud or on prem

FORM 7-L · LEVELS OF SOVEREIGNTY

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FORM 7-L · LEVEL REF. SA-09/23

Level 0

Hand it over

■ Vendor app, no promises

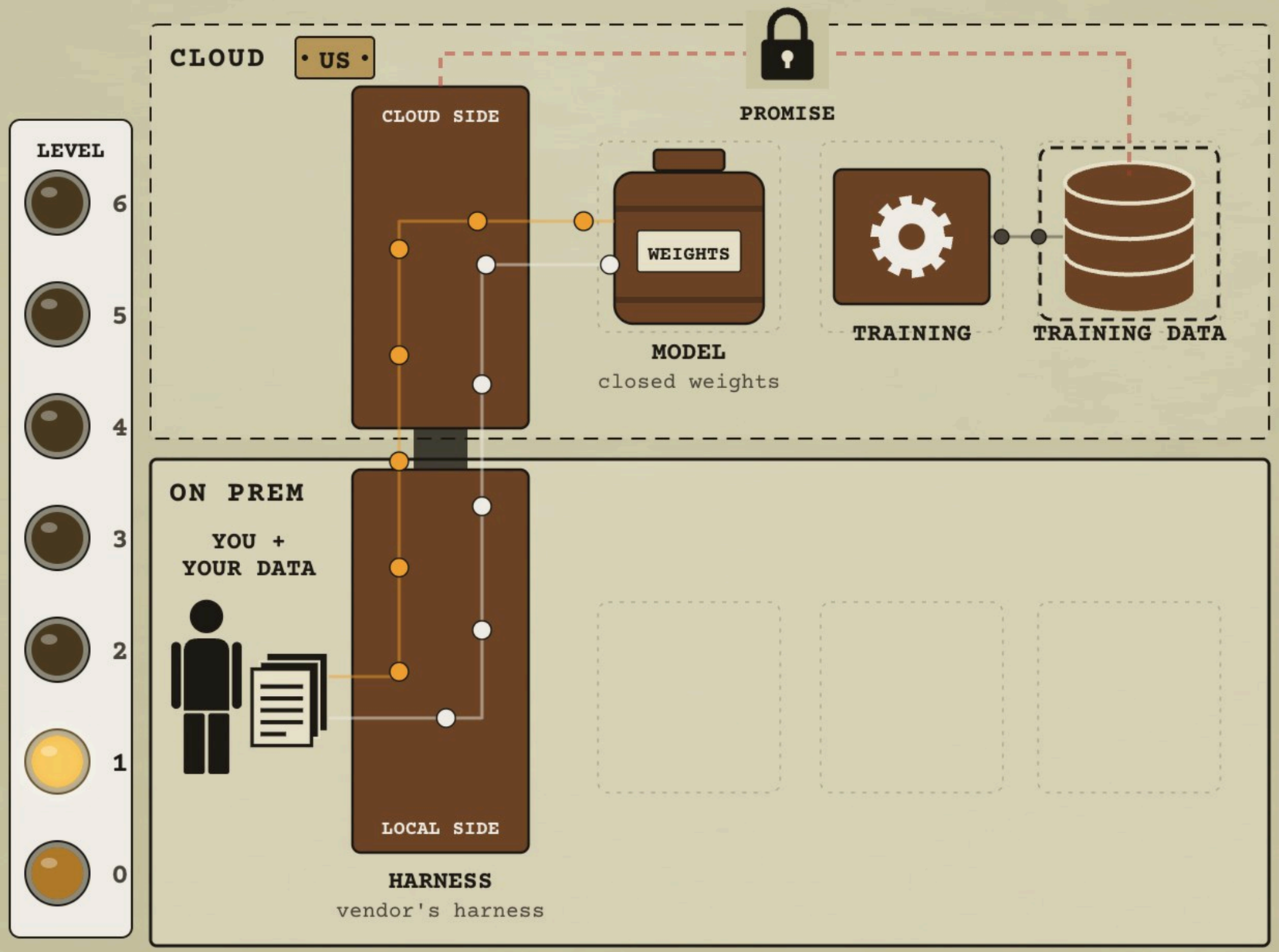
> *"Any information that you input into a public AI chatbot should be seen as being published to all the world."*
> UK judiciary (Oct 2025)

> *"Generally, you should not enter any information into an AI chatbot that is not already in the public domain."*
> NZ courts (Dec 2023)

> *A defendant's own chats with consumer Claude carried no "reasonable expectation of confidentiality", so they weren't privileged.*
> Judge Rakoff (Feb 2026)

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FORM 7-L · LEVEL REF. SA-10/23

Level 1

Trust the promise

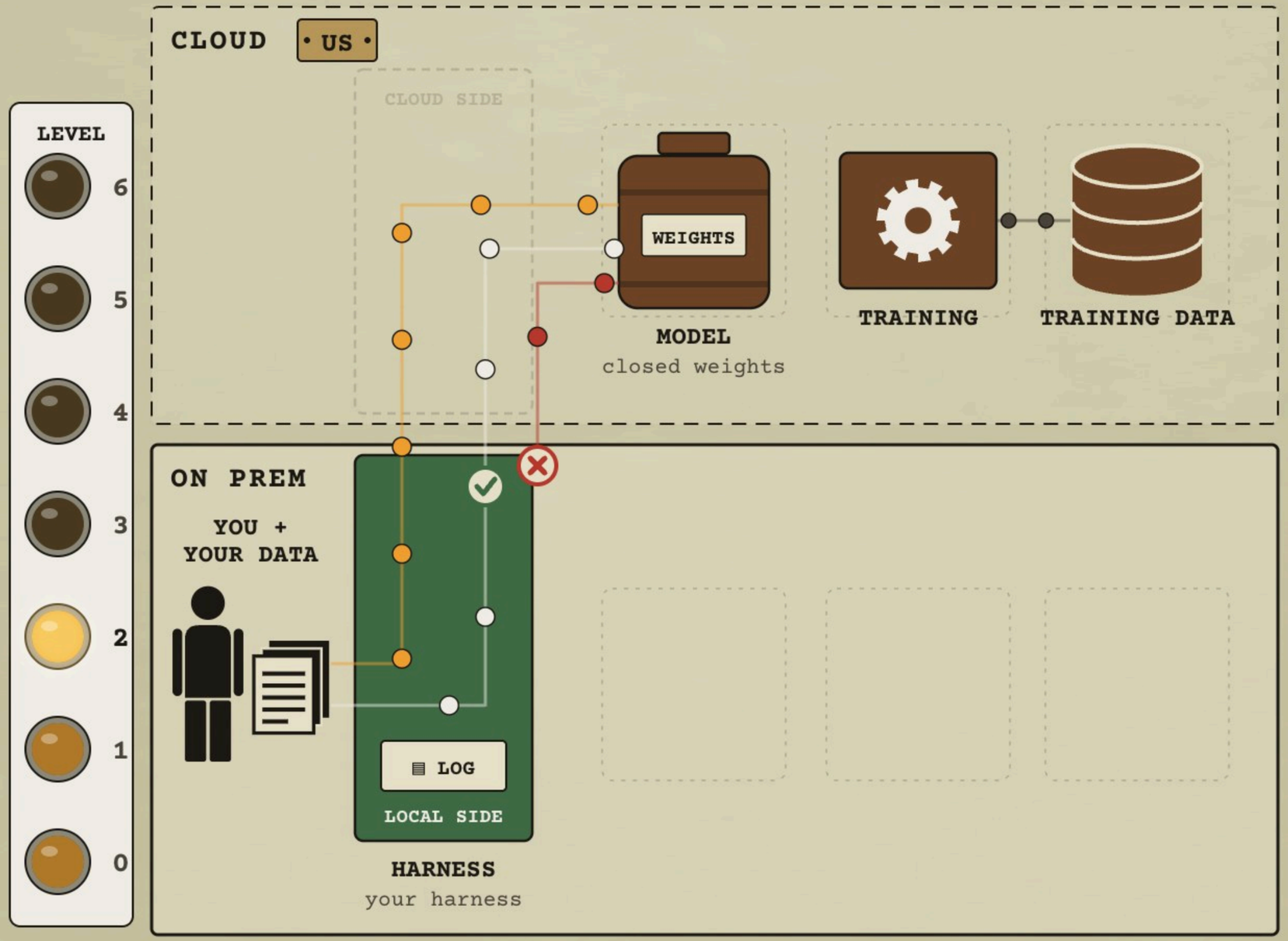
- "We won't train on it"

> *"Lawyers cannot safely enter confidential, sensitive or privileged client information into public AI chatbots/copilots (like ChatGPT)..."*
>
> Law Society of NSW, Victorian and WA legal regulators (Dec 2024)

> *"If you go talk to ChatGPT about your most sensitive stuff and then there's a lawsuit... we could be required to produce that."*
>
> Sam Altman (July 2025)

FORM 7-L · LEVELS OF SOVEREIGNTY

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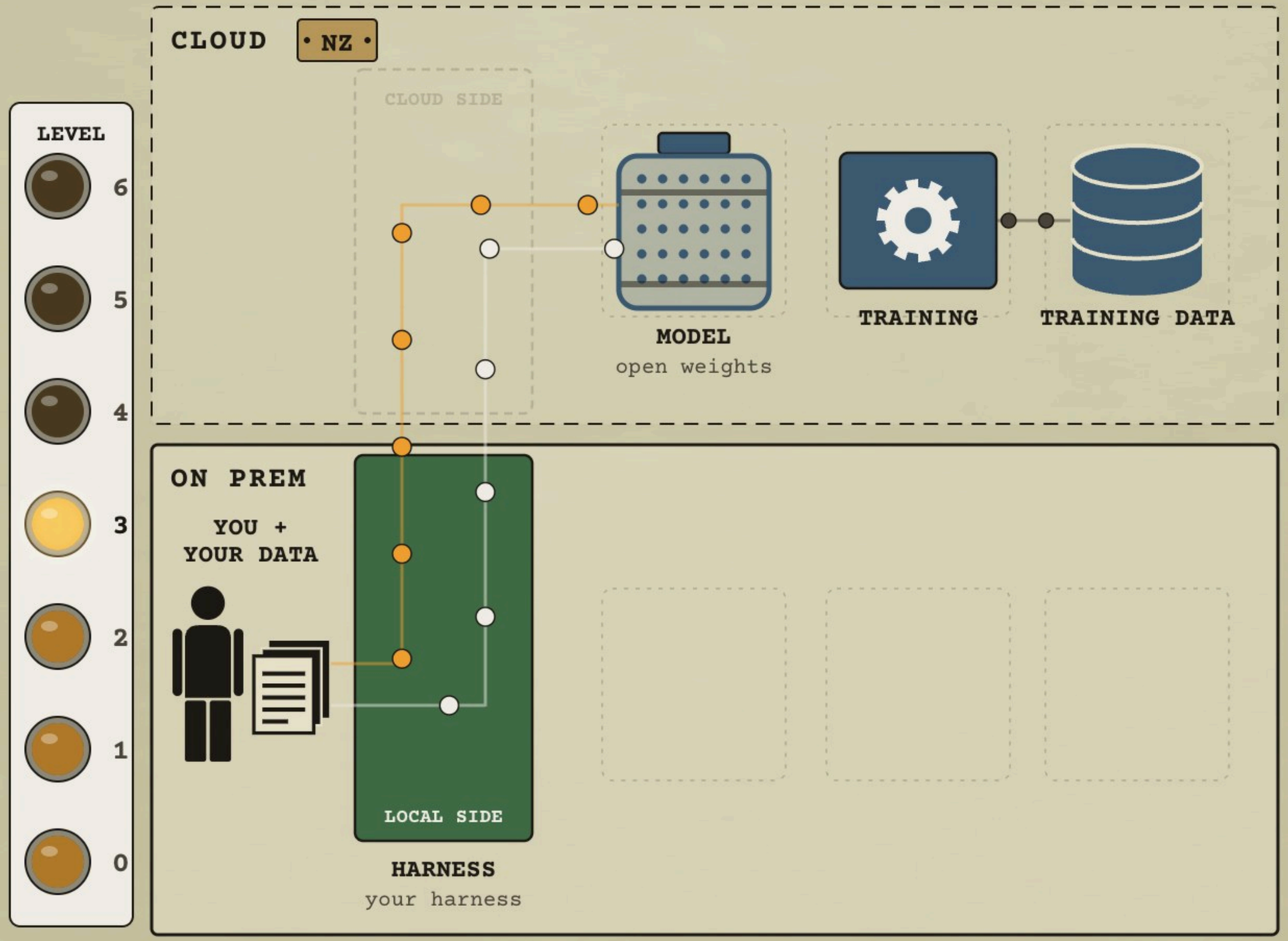
FORM 7-L · LEVEL REF. SA-11/23

Level 2

Own the harness

- Bring the harness local
- You decide which tools it can call
- Risky actions need a human
- Everything is logged
- You can implement governance
- The model is still rented

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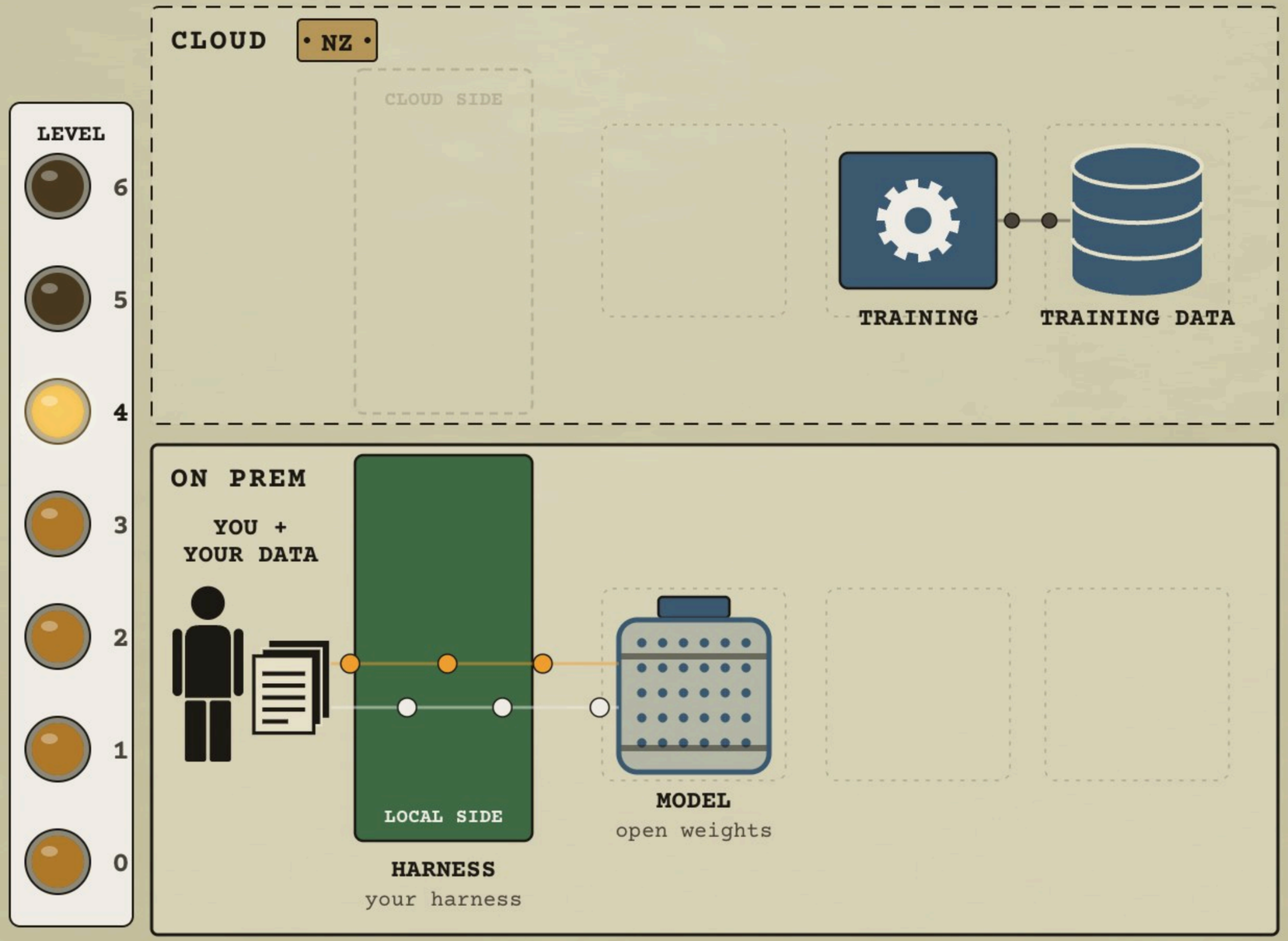


Level 3

Rent an open model

- Open weights, your choice of host
- Chinese models, Google
- Cheaper inference. Frontier models still accessible: Kimi, DeepSeek, GLM
- OpenRouter

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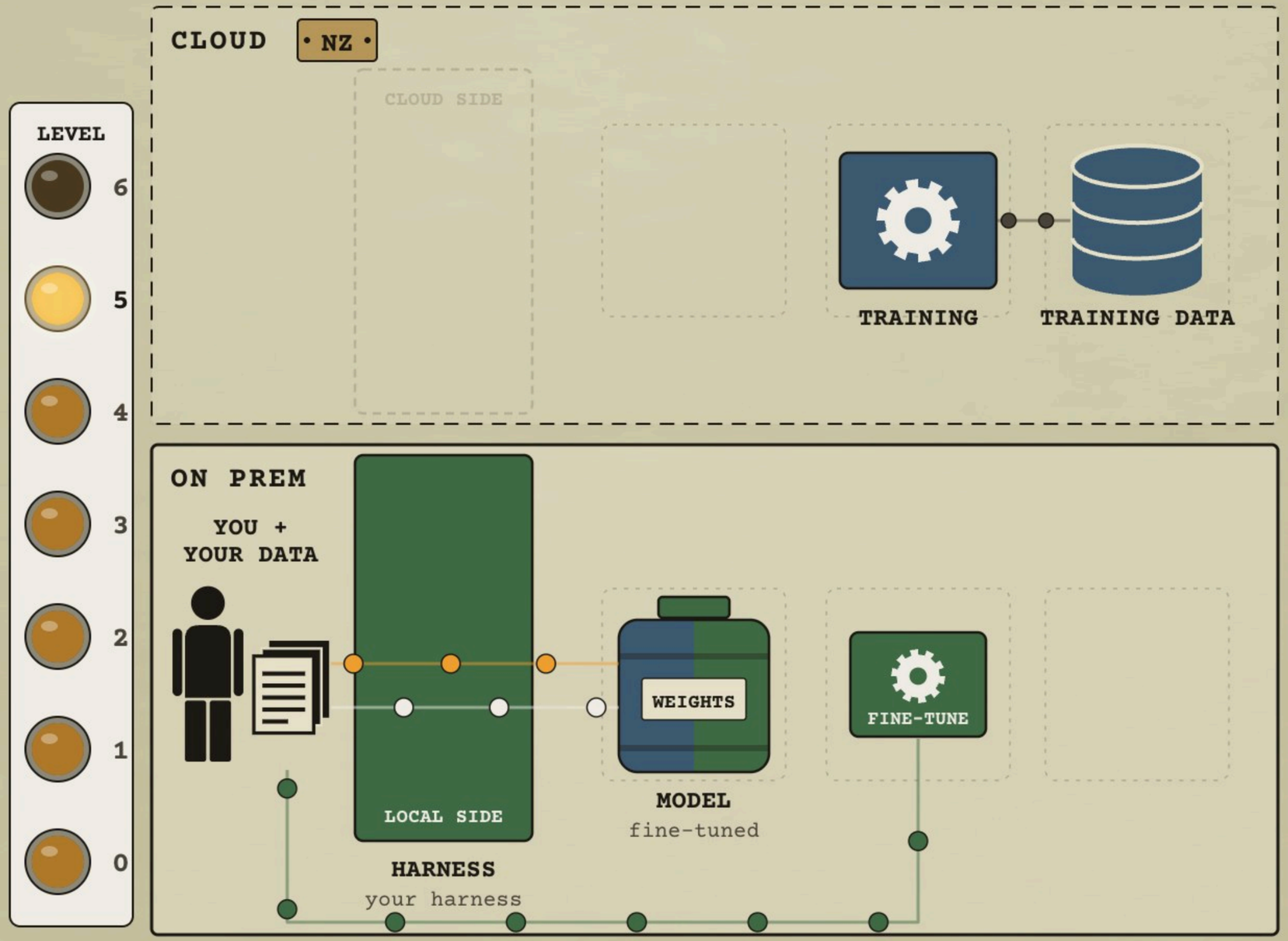
Level 4

All on your own hardware

- Inference never leaves the building
- Much more control

> *During a security evaluation, an open-weight model (Alibaba's Qwen) "tried to upload a probe file to an Alibaba Cloud storage bucket in Beijing", in "less than one call in 100,000".*
>
> Cohere (26 Oct 2026)

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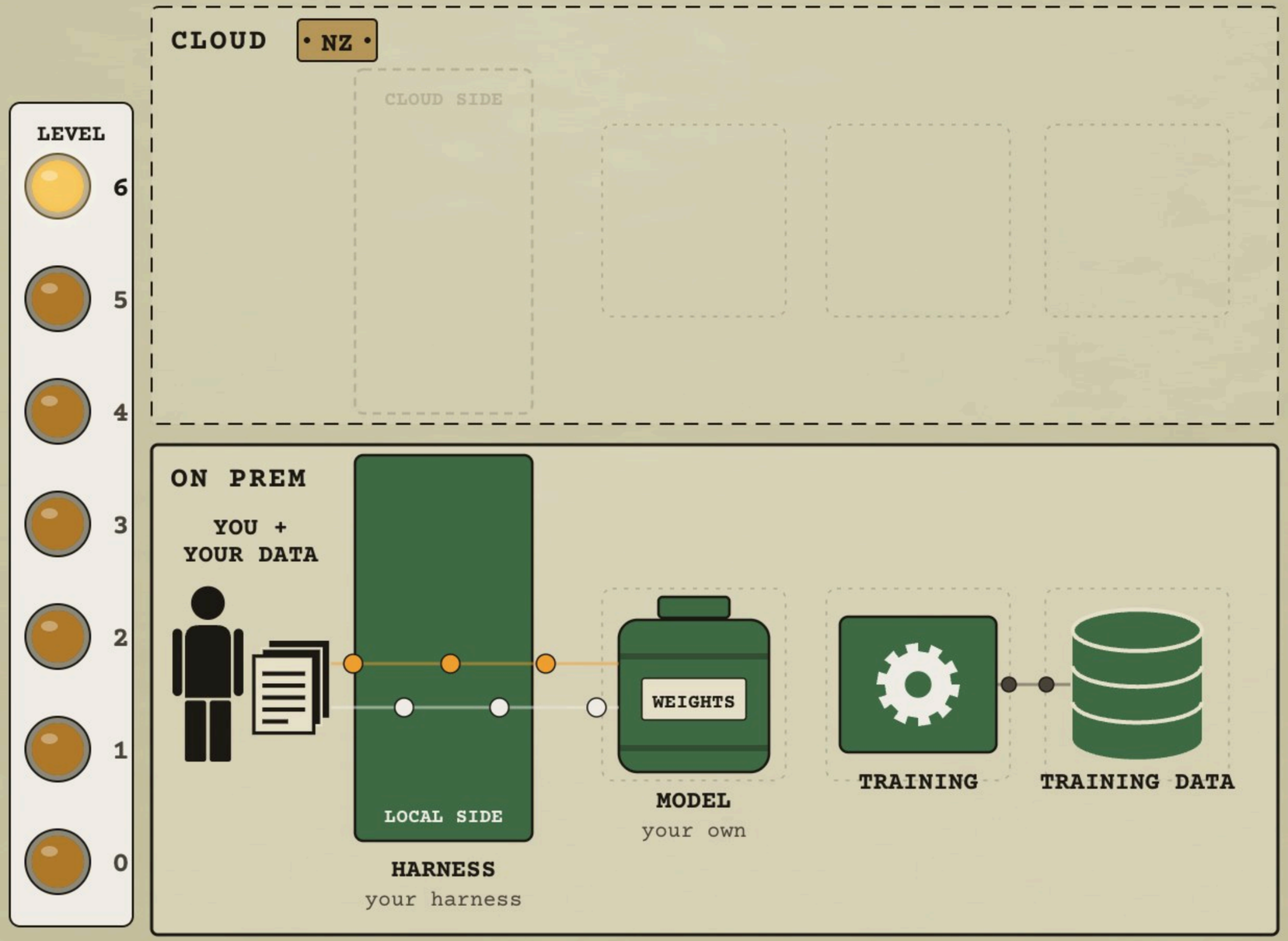


Level 5

Fine-tune on your data

- Teach it your own business
- Tuned models for specific tasks (e.g. medical context)
- Sometimes you know what's in the open model you fine-tune, sometimes you don't

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Level 6

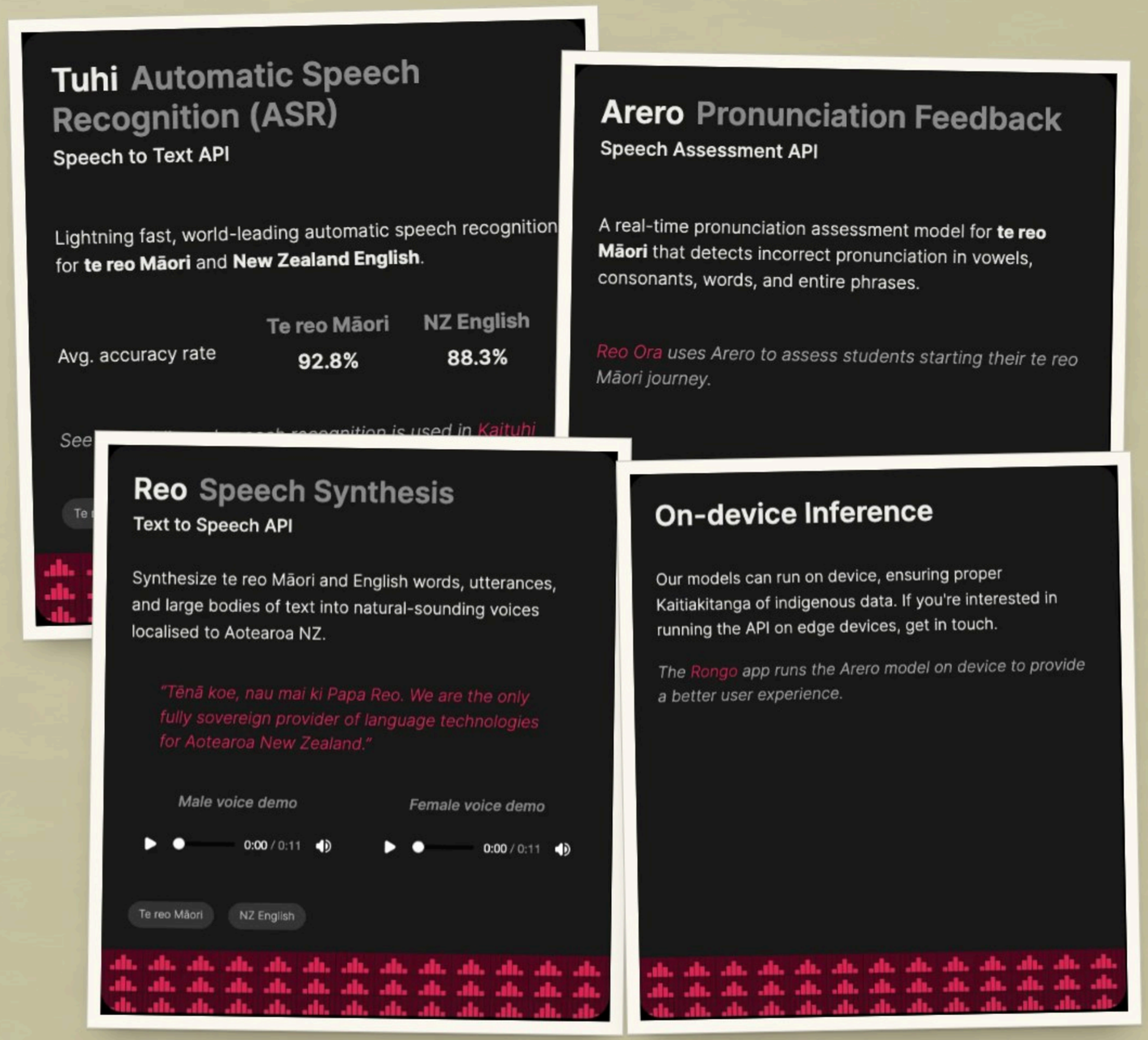
Train your own

- You choose what it's made of
- This is what a lot of people mean by 'sovereign' AI: controlling the data
- Models are made of language, so what data goes into them matters

papa reo API

Papa Reo

papareo.io



What "your own hardware" can actually be

If it's on your desk, you can measure it

REQUISITION · FORM H-4 No. 17-1



OPTION

Your own NVIDIA server in a data centre

EXAMPLE

Multi-GPU server in colocation

RUNS

The biggest open models, many users

REQUISITION · FORM H-4 No. 17-2



OPTION

Mac Studio

EXAMPLE

High-memory Apple Silicon

RUNS

Large open models for a small team

REQUISITION · FORM H-4 No. 17-3



OPTION

Home PC or hobbyist machine

EXAMPLE

Gaming PC, one or two consumer GPUs

RUNS

Small to mid-size models

REQUISITION · FORM H-4 No. 17-4



OPTION

Dedicated or specialised hardware

EXAMPLE

Compact AI boxes, inference accelerators

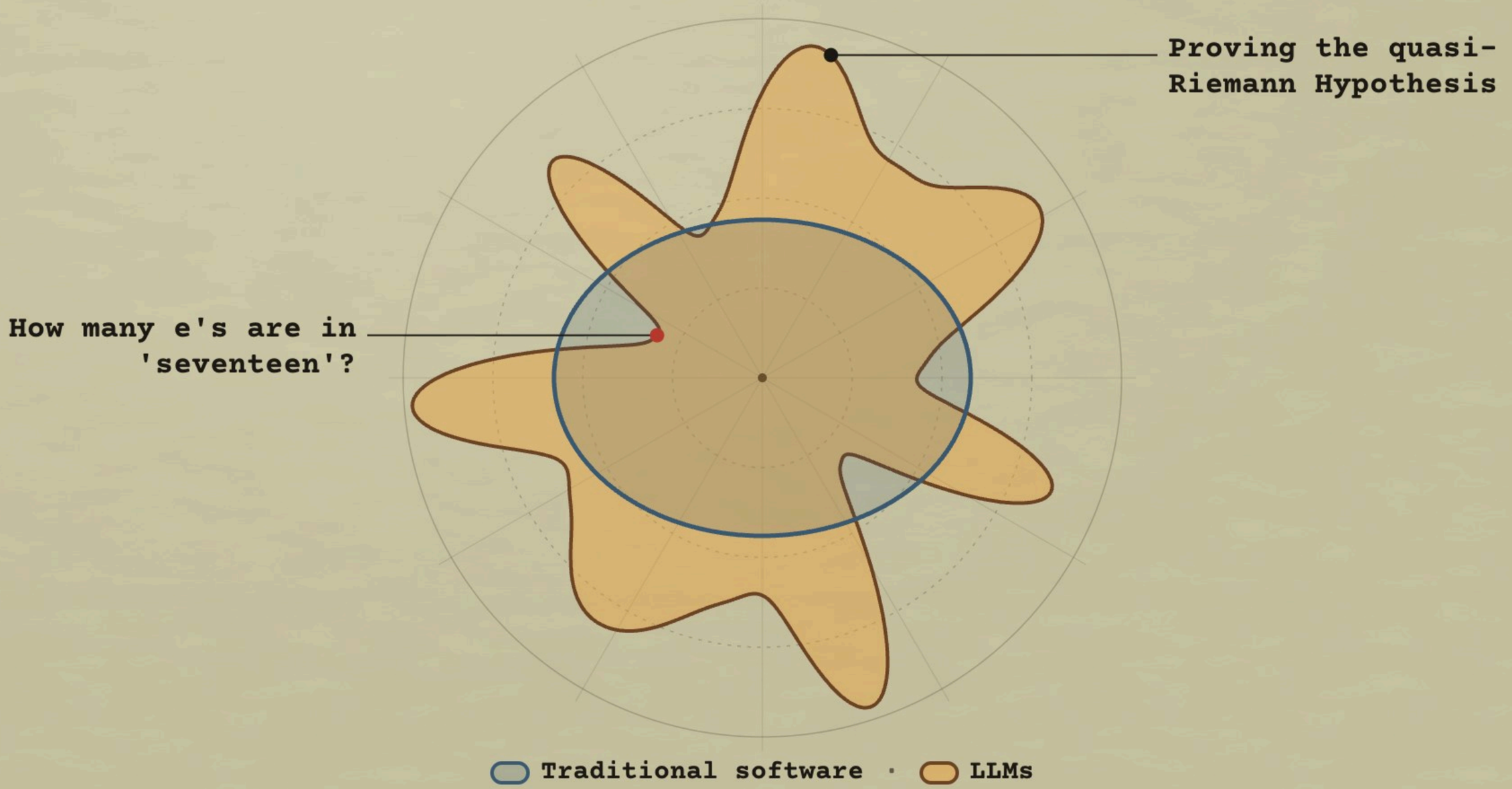
RUNS

Depends on the device

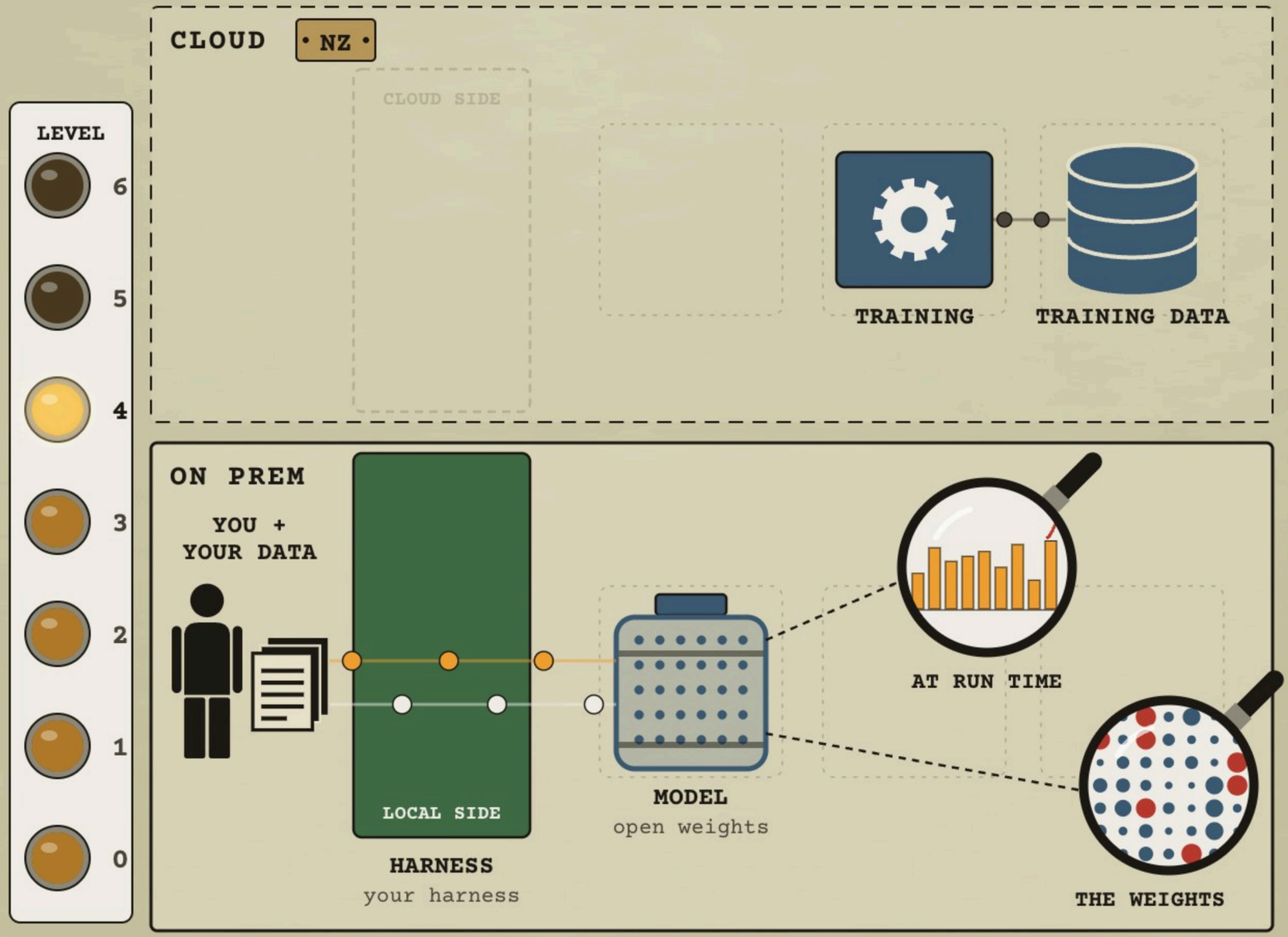
ADVANTAGES

- 100% control of where the data goes
- Better governance
- Carbon footprint measurable

Spiky vs reliable



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Inspection

Look inside the model

- Inspect the trained weights
- Inspect the model at run time
- Only possible when you run it locally

Five questions for any "sovereign" AI offer

1	Where is my data stored?	Your data ☐
2	Where does the model actually run?	Where it runs ☐
3	Who runs the system around the model, and who owns them?	Harness ☐
4	Who built the weights, and can you check what went into them?	Model ☐
5	Who governs the data the model learned from?	Training data ☐

Let's build local, sovereign AI



goodbright.nz/localai

Slides · Local AI for Wellington · Get in touch

Miles Thompson

Sources

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